

Build Your Own AI Assistant

1. Summary

This 5-session (2 hours each) hands-on workshop will guide students in building a personalised RAG-powered AI assistant. Each group will choose one theme including Health, Science, Sustainability, or Literature, and develop their system step by step. The workshop will start with LLM basics and data exploration, then move to document retrieval using vector databases and grounded answer generation with LLMs. Students will integrate these components into a complete RAG pipeline and may add simple personalisation features, such as adjusting the level of detail in responses. The final session will be dedicated to student presentations and live demos.

2. Workshop Outline

Session 1: Kick-off & Setup

- Intro to LLMs & RAG
- Overview of the project and themes
- Choose your theme
- Overview of tools and libraries (Python, LangChain, Hugging Face)
- Fetch and explore sample data for your theme

Session 2: Retrieval Basics

- Process and chunk your chosen data
- Build a simple vector database
- Create a basic retriever with LangChain
- Test retrieving relevant info for sample queries

Session 3: Generation & Fine-Tuning

- Intro to fine-tuning.
- Fine-tune a model (e.g. Mistral or BioGPT) on your theme's data
- Generate summaries or answers from retrieved content
- Quick evaluation (e.g., check quality manually or with simple metrics)

Session 4: RAG Integration & Personalisation

- Connect retrieval + fine-tuned LLM into one pipeline
- Add personalization (e.g., adjust detail level)
- Build and test a working prototype
- Simple demo / UI option (e.g., Streamlit)
- Fix common issues together

Session 5: Presentations & Wrap-Up

- Each group will present their AI assistant (10-15 min: demo + what they learned)
- Group feedback and discussion
- Next steps (advanced features, deployment, etc.)

3. Project Themes & Datasets

Theme	Datasets
Health	https://medlineplus.gov/xml.html
Science	https://www.kaggle.com/datasets/Cornell-University/arxiv/data
Sustainability	https://www.theccc.org.uk/wp-content/uploads/2025/06/Progress-in-reducing-emissions-2025-report-to-Parliament.pdf https://assets.publishing.service.gov.uk/media/642d85fcfbe62000c17dd71/Net_Zero_Society_Report_2023.pdf
Literature (Shakespeare's Works)	https://www.gutenberg.org/ebooks/100

4. Software Requirements

All workshop activities can be completed using *Google Colab*, with no local installation required. We will use open-source LLMs and publicly available datasets. Students will work with Python notebooks and core libraries such as *LangChain* for building RAG pipelines, *Hugging Face Transformers* for accessing LLMs, and lightweight vector databases (e.g. FAISS or Chroma) for retrieval. Students will only need a Google account and a web browser.